

# **Bone Disorders:**

## **Osteoporosis Management and Complications**

### **Medication-Related Osteonecrosis of the Jaw**

**Integrated Biomedical and Clinical Sciences OMFS513**

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Mission: To advance the science and art of health through  
education, service, scholarship and social accountability

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## Learning Objectives

### Learning Outcomes

1. Describe hereditary and metabolic bones disorders and joint disorders.
2. Describe the pathophysiology of osteoporosis
3. Describe bisphosphonate classes, indication and adverse effects.
4. Understand the risk factors leading to MRONJ development and its management strategies.

## Outline

Hereditary and metabolic joint & bone disorders

Osteoporosis

Bisphosphonate classes, mechanism of action and adverse effects.

What is MRONJ?

MRONJ staging and management.

## Joint Disorders

### 1- Gout:

Acute inflammation due to the deposition of monosodium urate (MSU) crystals (due to excessive nucleotide turnover or decreased excretion).

**Risk factors:** male sex, genetic predisposition, diet, hypertension, diabetes, kidney disease and medications like thiazide diuretics and aspirin.

### Pathophysiology

MSU crystals in synovial fluid triggers release of interleukins and initiation of inflammatory process. Prolonged accumulation results in joint damage “gouty arthritis”.

**Symptoms:** Inflammation, pain and joint swelling (common in the first toe). Chronic gout attacks may lead to ongoing synovitis, deformity, and bony destruction.

**Diagnosis:** Identification of MSU crystals. Labs and clinical.

**Treatment:** Acute flare: NSAIDs and steroids to control inflammation. Long term: urate lowering therapy like allopurinol



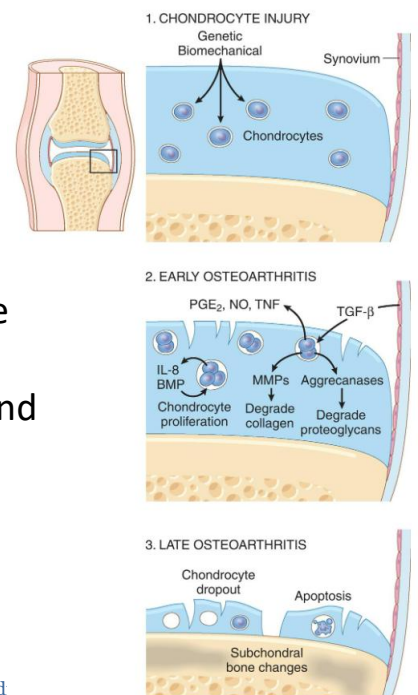
**Figure 111.4** Gout of the great toe presenting as erythema of the overlying skin.

## 2- Osteoarthritis

Most common form of age-related joint disease involving the synovial joints.

**Causes:** degenerative joint disorder that cause the destruction of the articular cartilage eventually leading to joint damage. Common in knee, hip, hand and spine joints.

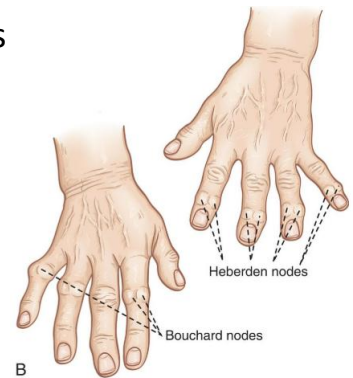
**Risk factors:** Female, older age (>60), obesity and joint trauma.



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### Symptoms:

- Pain and burning sensation in weight-bearing joints that gets worse with activity and improves with rest.
- Mild and short morning stiffness (around 15 min), usually no joint swelling.
- Motor restriction
- Joint noises or grinding sounds may be associated with movement.



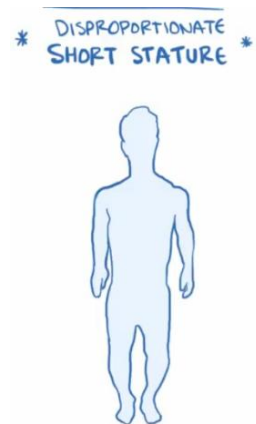
**Treatment:** NSAIDs, intra-articular glucocorticoids, lifestyle and surgery.

## Skeletal Disorders

### Achondroplasia

**Prevalence:** most common type of inherited dwarfism (1:20,000).

**Etiology:** caused by an autosomal dominant (gain of function mutation) in fibroblast growth factor receptor 3 (FGFR3) which inhibits cartilage synthesis at the epiphyseal growth plate and decrease bone formation.





### Symptoms:

- Disproportionate dwarfism (short and thick extremities) and macrocephaly.
- Bowed lower legs, and trident hand (extra space between the middle and ring fingers).
- Flat bones (skull, trunk) are usually normal leading to relatively large head and trunk.
- Crowded/misaligned teeth, and maxillary hypoplasia.



## Osteopetrosis “Marble bone disease”

**Causes:** Hereditary, leading to decreased osteoclast function and decreased bone resorption. Results in dense, thickened but defective bones (stone bones) that can fracture easily.

### Types:

#### **1- Autosomal recessive “Infantile malignant type” .**

Caused by mutations in carbonic anhydrase 2 (CA2) gene which impairs the ability of osteoclasts to generate acidic environment.

Symptoms: Impaired growth, skeletal sclerosis (abnormal hardening), bones fracture easily, bone marrow failure (pancytopenia), vision and hearing impairment, delay in tooth eruption, enamel hypoplasia, caries.

## 2- Autosomal dominant “Adult benign type”

(aka, osteopetrosis tarda):

Caused by mutations in chloride channel 7 (*CLCN7*) gene, less severe type.

Symptoms develop at adult stage ranging from asymptomatic to pathological fractures, vision/hearing loss.

### Management:

- **Congenital form:** Bone marrow transplant and Supportive: calcium, vit. D supplementation, erythropoietin
- **Osteopetrosis Tarda:** Supportive to repair fractures, physical therapy...

## Osteogenesis Imperfecta

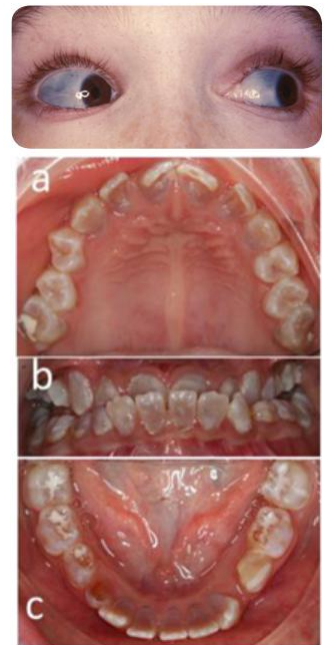
**Causes:** autosomal dominant mutations genes encoding  $\alpha 1$  and  $\alpha 2$  chains of type I collagen (example: *COL1A1* and *COL1A2*) leading to defective collagen type I synthesis.

**Types:** Nine subtypes with different severity (type I is the mildest, type II fatal, III-IX variable symptoms) .

**Symptoms:** Brittle bone (osteopenia) leading to recurrent fractures and skeletal deformity, blue discoloration for the sclera, hearing loss.

In some cases, dentinogenesis imperfecta: small, fragile, and discolored teeth (due to dentin deficiency) was reported.

**Treatment:** bisphosphonates, surgical, physical therapy.



16-y-old female with type III OI.

[Osteogenesis imperfecta tooth level phenotype analysis: Cross-sectional study - PubMed \(nih.gov\)](#) California Northstate University College of Dental Medicine

## Paget's Disease

Increased bone remodeling of undetermined cause. Affects adults >50 Y. 2nd most prevalent metabolic bone disorder (1.5%-8.3%). Can affect one or multiple bones, most often in axial skeleton.

### Pathophysiology: Three distinct phases:

- 1- lytic phase: Increased osteoclast activity, causing lytic lesions.
- 2- Mixed: osteoclast activity with highly increased osteoblastic activity that produces a disorganized bone, yielding a deformed bone.
- 3- osteosclerotic phase: osteoblast activity is predominant leading to disorganized, thicker and weaker bone.

**Symptoms:** initially asymptomatic. Symptoms depend on severity and stage: bone pain, increase risk of fractures, bone deformation and enlarged skull and sometimes enlarged jaw bones.

**Management:** bisphosphonates and calcitonin



A 65-year-old female patient Paget's disease in the mandible

## Osteomalacia/rickets

**Causes:** Bone softening disease.

Metabolic bone disease due to defective mineralization of bone matrix or cartilaginous growth plates (rickets, only children).

Most commonly secondary to vit D deficiency. Decreased vit D will decrease serum calcium, increase PTH and decrease phosphate level.

**Symptoms:**

- In children (rickets), softening of the bones, impaired bone growth, altered gait, bowlegs and bone malformations.
- In adults, osteomalacia will cause bone fragility, increase risk of fracture or falls and bone/joint pain. Rickets is associated with abscessed non-carries tooth.

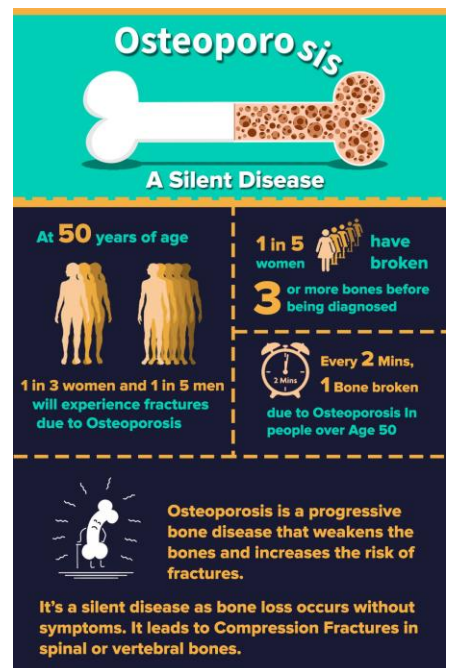


**Treatment:** Vit D and correction of calcium intake

## Osteoporosis

Most common bone disorder.  
Osteoporosis is a progressive metabolic bone disorder.

WHO definition: “a systematic skeletal disease characterized by low bone density and microarchitectural deterioration of bone tissue with a consequent increase in bone fragility.”

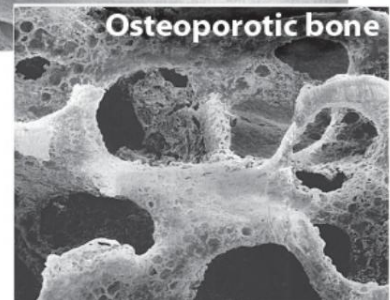
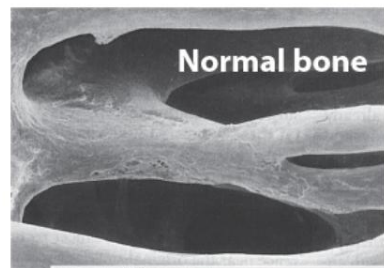


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Osteoporosis is characterized by:

- Loss of bone matrix (tubercular bone).
- Thinning of the cortical bone
- Increased risk of bone fractures due to osteoclastic function.
- Can be localized or generalized (involving the entire skeleton).



**Prevalence:** Around 19.5 % for females (>50 Y) and 4.5 % in males (>50 Y). Causes over 1.5 million fractures annually. Most commonly vertebral fractures, followed by hip fractures (300,000 annually) and pelvic fractures.

**Risk factors:** female sex, older age, non-Hispanic white and Asian, menopause, genetic factors, lack of physical activity (in contrast, weight-bearing exercises increase the bone mass) and nutrition.

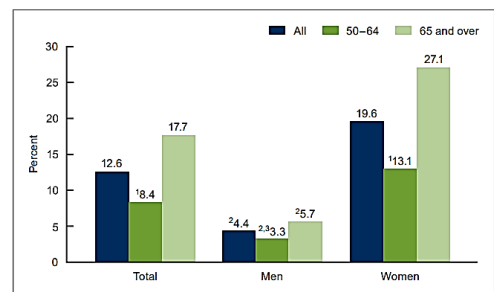
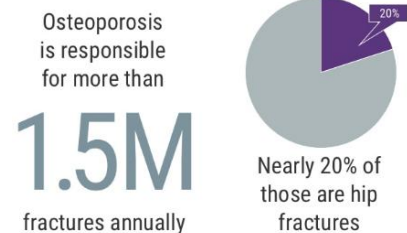
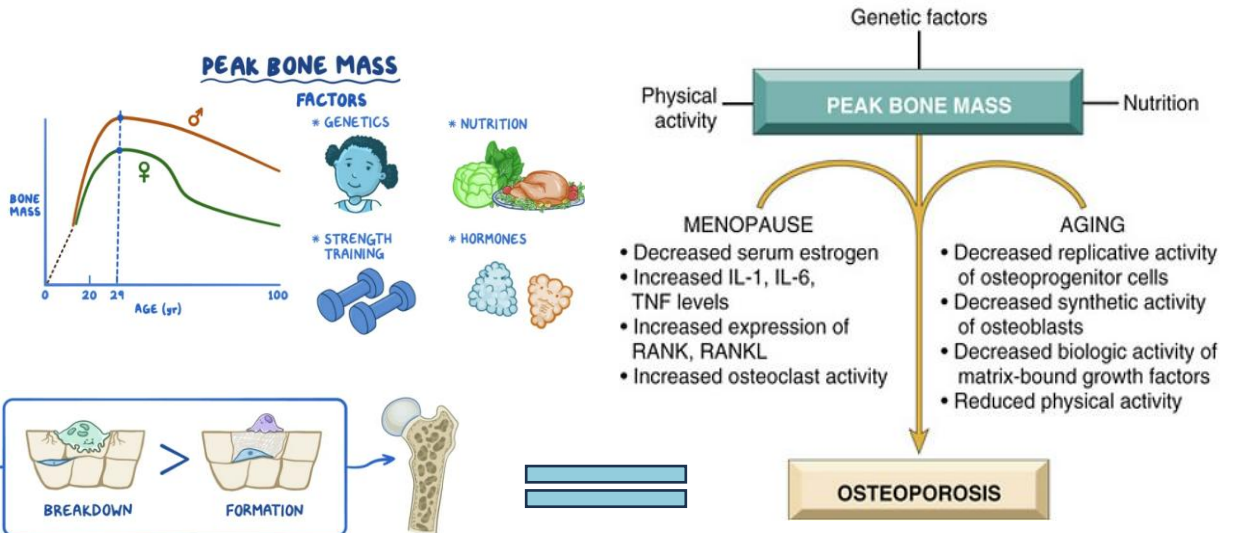


Figure 1. Prevalence of osteoporosis among adults aged 50 and over, by sex and age: United States, 2017–2018



Source: Global health estimates, World Health Organization, July 2013

# Pathophysiology



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## Common Types

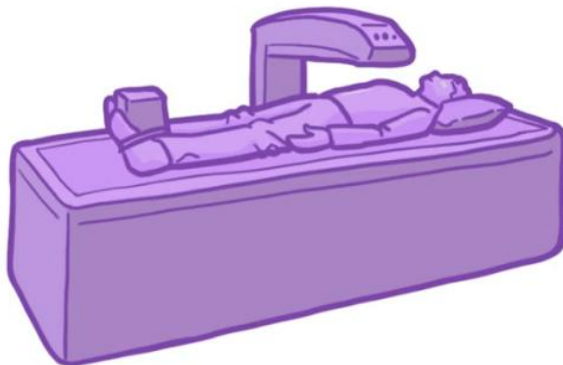
- Menopausal osteoporosis: Around 40% of postmenopausal females. After menopause, decreased estrogen levels increase bone resorption. Decreased estrogen and/or menopause-related events can increase the inflammatory cytokines (IL-6, TNF- $\alpha$ , and IL-1). Increase RANKL and decrease OPG leading to increased osteoclast action and higher bone resorption.
- Senile osteoporosis: with advanced age ( $\geq 70$ ), male and females. Bone marrow cells differentiate into more adipocytes rather than osteoblasts. The osteoblasts have reduced proliferative potential and reduced response to growth factors compared to osteoblasts in younger individuals. The net result is a diminished capacity to make bone (senile bone loss is around 0.7 % /year).

## Less common types: Osteoporosis secondary to:

- **Medications:** like glucocorticoids, thyroxin, and others
  - ❖ Glucocorticoids decrease serum calcium via decreasing calcium absorption from the GI and increasing renal excretion of calcium, increase osteoclast survival. Prednisone 5 mg/d (or equivalent) for  $\geq 3$  months increase risk for osteoporosis.
  - ❖ Long-term use of thyroxin may decrease bone mineralization density and increase the risk of fractures. Mechanism is not understood.
  - ❖ Heparin can enhance the osteoclastic activity via inhibiting OPG. Mechanism is not completely understood.
  - ❖ long term use of PPIs ( $\downarrow \text{Ca}^{+2}$  absorption), anticonvulsants ( $\uparrow$  CYP450 which metabolize vit D) and others
- **Others:** Hyperparathyroidism (PTH), diabetes, malabsorption diseases (e.g celiac disease), alcoholism, reduced physical activity (bed-bound patients).

## DIAGNOSIS

### DUAL-ENERGY X-RAY ABSORPTIOMETRY (DEXA) SCAN



#### \* TESTS BONE DENSITY

INDIVIDUAL'S vs. NORMAL ADULT

T SCORE  
 $\leq -2.5$

OSTEOPOROSIS

**WHO criteria:** bone mineral density (BMD)  $\leq -2.5$  standard deviations than average BMD value for young healthy person (T-score  $\leq -2.5$ ) = Osteoporosis.

Other Values: Normal: T score  $> -1.0$ ; Osteopenia: T score  $-1.0$  to  $-2.5$

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## Presentation

- Silent or asymptomatic until fractures occur.
- Back pain.
- Spontaneous fracture or fracture with minor falls. Frequently in vertebra (thoracic & lumbar) or weight bearing like femur neck or pelvis leading to mobility complications.
- Spinal kyphosis
- Loss of height (common)



## Management and Prevention

- Proper nutrition
- Exercise “Mechanical stress/forces stimulate normal bone remodeling (e.g weight training is better than cycling)”
- Lifestyle changes (smoking cessation, limit alcohol intake)
- Fall prevention to help prevent fractures
- Pharmacological treatment



### OSTEOPOROSIS PREVENTION



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# Osteoporosis Treatment

## Non hormonal treatment

Bisphosphonates

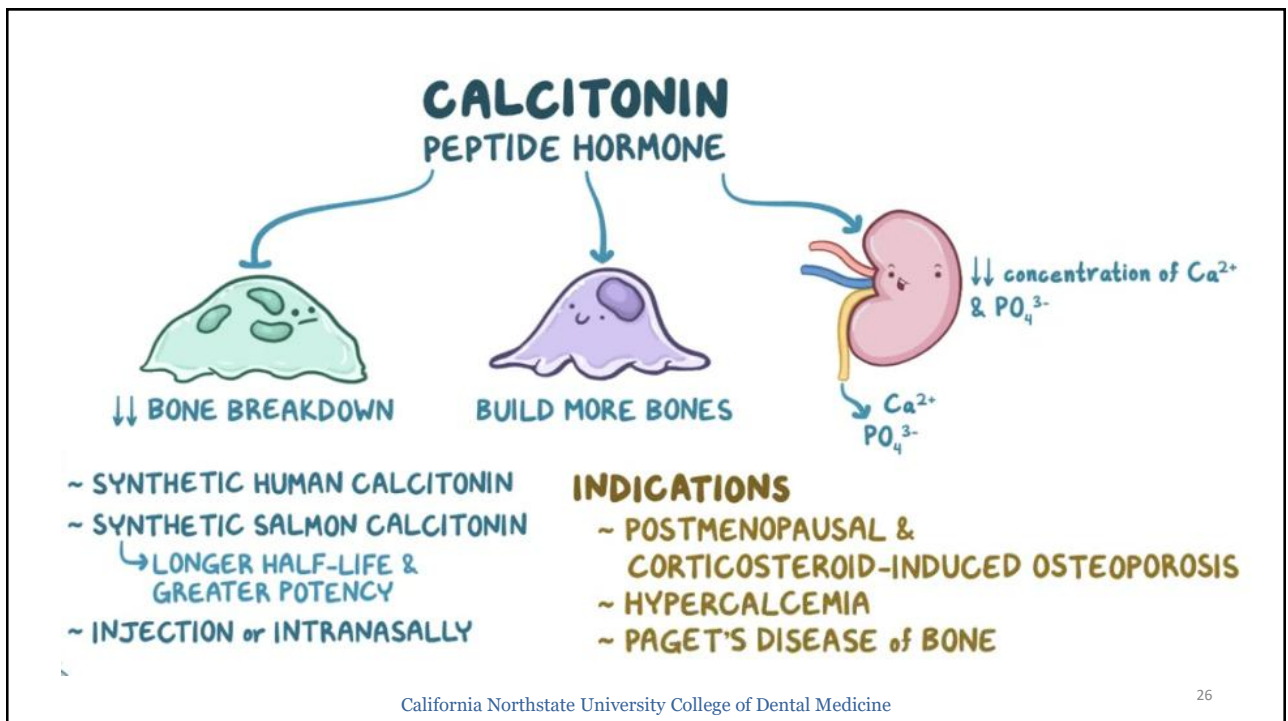
denosumab

## Hormonal treatment:

Calcitonin: Reduces the bone resorption

Raloxifene: a selective estrogen receptor modulator.

Teriparatide: recombinant form of human parathyroid hormone.



## TERIPARATIDE

~ RECOMBINANT ANALOG of the PARATHYROID HORMONE

~ given SUBCUTANEOUSLY ONCE/DAY

STIMULATES OSTEOBLASTS



REDUCES APOPTOSIS

BUILD more BONES than what is  
BEING LOST



### SIDE EFFECTS

- ~ TRANSIENT HYPERCALCEMIA
- ~ ↑↑ RISK of developing OSTEOSARCOMA

## RALOXIFENE

~ selective estrogen receptor modulator (SERM)

~ PARTIAL AGONIST → ESTROGEN RECEPTORS in BONES

~ ANTAGONIST → ESTROGEN RECEPTORS in BREAST TISSUE

### INDICATIONS

- ~ OSTEOPOROSIS in POSTMENOPAUSAL INDIVIDUALS
- ~ PREVENTION of BREAST CANCER

### SIDE EFFECTS

- ~ HOT FLASHES

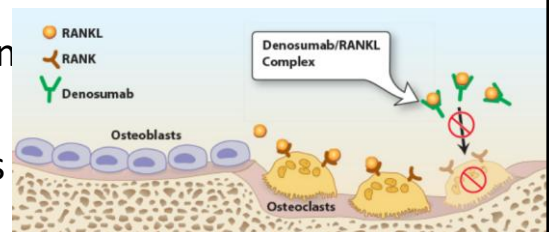
## Non-hormonal treatment

**Generic/Trade:** Denosumab (Prolia®).

**MOA:** monoclonal antibody against the RANKL. Reduces osteoclast differentiation and function.

**Indication:** postmenopausal osteoporosis in women at high risk of fracture. SC injection once/6 months.

**Side effects:** hypocalcemia, MRONJ and increased risk of infections.



## Bisphosphonates

### **SIMPLE, NON-NITROGENOUS BISPHOSPHONATES**

- ~ ETIDRONATE
- ~ TILUDRONATE

### **POTENT, NITROGENOUS BISPHOSPHONATES**

- ~ ALENDRONATE
- ~ IBANDRONATE
- ~ PAMIDRONATE
- ~ RISEDRONATE
- ~ ZOLEDRONATE

## Non-nitrogenous

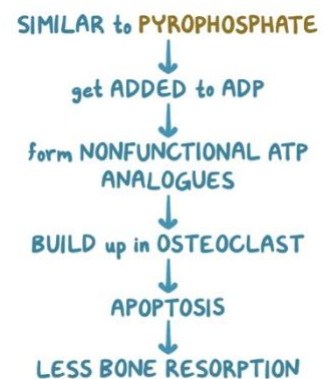
**Example:** Etidronate, Tiludronate (Skelid)

**Indication:** Treatment of Paget disease of the bone.

**MOA:** Cause osteoclast apoptosis and decrease bone resorption.

**Side effects:** MRONJ, Abnormal taste sensation.

*Both drugs were recently discontinued.*



Not in the exam

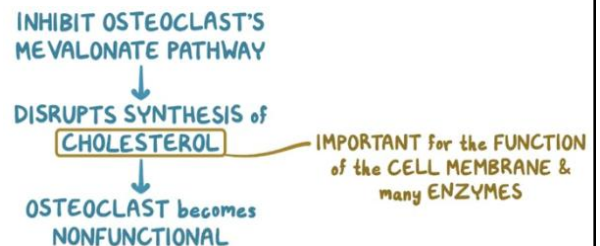
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## Nitrogenous Bisphosphonates

**Example:** alendronate (Fosamax®), risedronate (Atelvia®), and zoledronate (Zometa®).

**MOA:** bisphosphonates bind to hydroxyapatite crystals in the bone and decrease osteoclastic bone resorption, resulting in an increase in bone mass and decreased risk of fractures in patients with osteoporosis.



## Bisphosphonate Pharmacokinetics

| Generic name | Trade name | Route of administration |
|--------------|------------|-------------------------|
| Pamidronate  | Aredia     | IV infusion             |
| Alendronate  | Fosamax    | Oral                    |
| Risedronate  | Atelvia    | Oral                    |
| Ibandronate  | Bonvia     | Oral or parenteral      |
| Zoledronate  | Zometa     | Parenteral              |

| Bisphosphonate         | Antiresorptive activity |
|------------------------|-------------------------|
| <i>Etidronate</i>      | 1                       |
| <i>Tiludronate</i>     | 10                      |
| <i>Pamidronate</i>     | 100                     |
| <i>Alendronate</i>     | 1000                    |
| <i>Risedronate</i>     | 5000                    |
| <i>Ibandronate</i>     | 10,000                  |
| <i>Zoledronic acid</i> | 10,000                  |

For illustration

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## Pharmacokinetics

- Oral bisphosphonates are poorly absorbed after administration.
- Medication is administered on empty stomach with plenty of water. Patients should remain in an upright position for 30 min to decrease GI side effect.
- Intravenous bisphosphonate achieve 40 % bioavailability.
- Renally eliminated. Avoided in patients with severe renal impairment
- Following absorption, bisphosphonates are rapidly up taken by the bones where they bind to the hydroxyapatite. Once bound to bone matrix, they are cleared at a very slow rate (for example, alendronate has an elimination half life of 10 years).



## Bisphosphonates (cont.)

### Indications

- Osteoporosis.
- Paget's disease.
- Metastatic bone diseases (parenteral bisphosphonate therapy).
- Osteogenesis imperfecta

## Bisphosphonates (cont.)

### Adverse effects:

- GI disturbance and esophagitis.
- Fever, flu-like syndrome.
- Hypocalcemia.
- Medication-related osteonecrosis of the jaw (MRONJ). Risk of MRONJ increase with high drug dosage, long treatment interval, IV route and potency of medication.

## BRONJ, MRONJ, ONJ, BON

2002: First reports (months after Zometa was approved for IV use).

2005: FDA issued a warning for the bisphosphonate drug class.

Osteonecrosis associated with drugs was named by multiple terms including

- bisphosphonate-related osteonecrosis (BRON),
- bisphosphonate osteonecrosis (BON)
- bisphosphonate-related MRONJ (BRMRONJ)
- Medication related osteonecrosis of the jaw (MRONJ).

## What is MRONJ?

MRONJ is uncommon, but potentially serious condition which causes necrosis of the jaw bones in patients treated with bisphosphonates, denosumab, and other drug classes.

Manifest as: intraoral exposed bone, delayed healing, inflammation, or pain, and swelling especially following tooth extraction.



“Exposed necrotic bone after tooth extractions in a patient treated with i.v. zoledronic acid.”

[Bisphosphonate Associated Osteonecrosis of the Jaw: An Update on Pathophysiology, Risk Factors, and Treatment \(hindawi.com\)](#)

[Medication-related osteonecrosis of the jaw: definition and best practice for prevention, diagnosis, and treatment - Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology \(Medication-related osteonecrosis of the jaw in patients with cancer - UpToDate \(cnsu.edu\)\)](#)

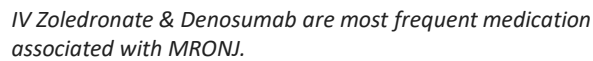
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Patients are considered to have MRONJ if they fulfilled the following criteria:

- Current or previous treatment with an osteoclast inhibitor or antiangiogenic agent. More than 90% of MRONJ cases occur in cancer patients receiving high doses of IV bisphosphonates or SC denosumab (120 mg every 4 weeks) therapy.
- Exposed or necrotic bone in the mandible or maxilla that has persisted for more than 8 weeks.
- No history of radiation therapy.

In the presence of a prior radiation therapy, the patient is categorized as an osteoradionecrosis (ORN) patient.



[Medication-related osteonecrosis of the jaw in patients with cancer - UpToDate \(cnsu.edu\)](#)

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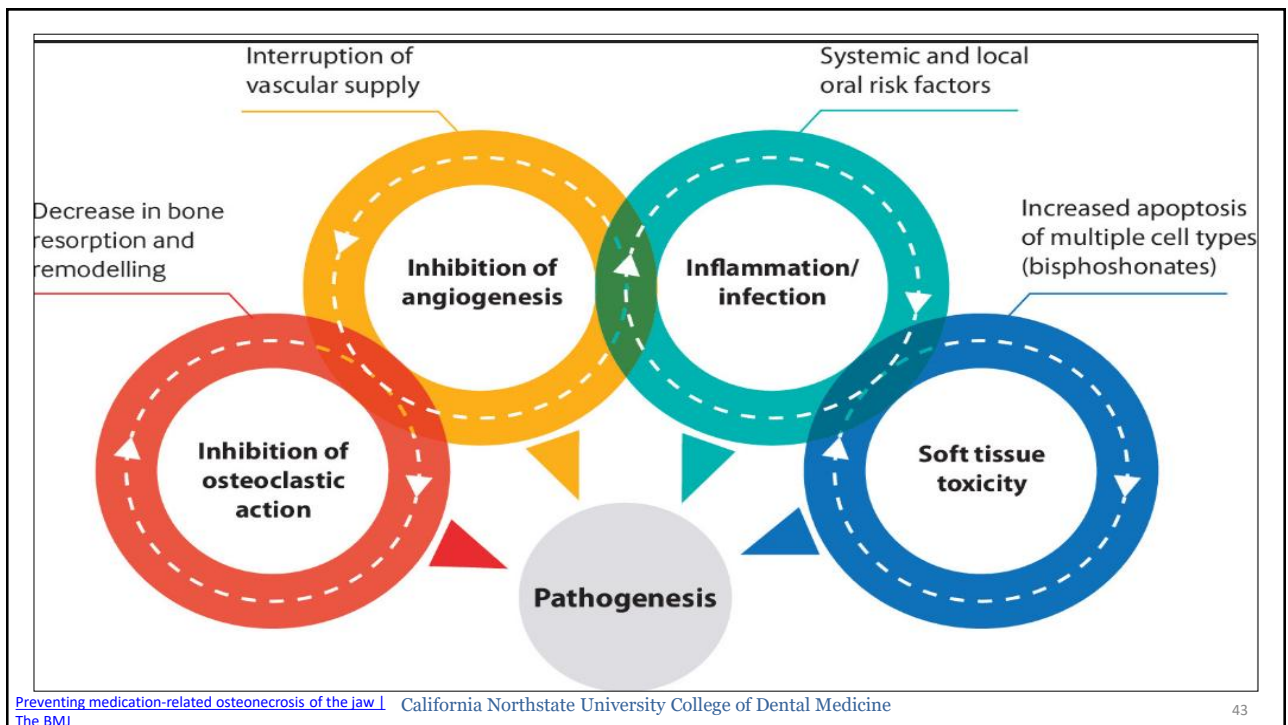
- **Oral health-associated risk factors:**
  - invasive dental procedures (up to 61% of the patients reported tooth extraction as the causative factor).
  - trauma, pre-existing inflammatory dental diseases like periodontal diseases and poor oral hygiene. Dentures can increase the risk of jaw osteonecrosis.
  - MRONJ is more likely to appear in the mandible (73 %) than the maxilla (23 %) .
- **Tumors**, especially myeloma.
  - Evidence demonstrated higher risk of MRONJ in myeloma patients. It was also seen in with other types of tumors (renal, breast and prostate).



- **Genetic factors:** Mutations in gene regions associated with bone turnover, collagen formation, or certain metabolic bone diseases are associated with higher risk of MRONJ.
  - For example, polymorphisms in the *RBMS3* gene causes a sixfold increase in MRONJ risk.
- **Other comorbidities,** like anemia, diabetes mellitus, and renal failure.

## Why the Jaw?

- a) Alveolar bone turnover is significantly higher than many other bones.  
Therefore, the alveolar bones may incorporate more bisphosphonate than the other skeletal bones.
- b) Jaw is one of the least-protected bones from infection in the human skeleton due to its proximity to the oral microflora.
- c) Jaw is subjected to mechanical stress due to the mastication force.
- d) Embryonic origin of the jaw bones is different from the other bones.



## Prevention

**1- Pretherapy:** Dental exam and screening. nation. High-risk invasive dental procedure can be performed.

**2- During antiresorptive therapy:** Follow up. Patient education. Discuss recommended oral hygiene instructions.

**3- Emergency:** Coordination of care between the health care providers. Surgical/conservative techniques. Drug Holiday (controversial) .

[American Association of Oral and Maxillofacial Surgeons' Position Paper on Medication-Related](#)

[Osteonecrosis of the Jaws-2022 Update - PubMed \(nih.gov\)](#)

[The effectiveness of different treatment methods in the treatment and prevention of medication-related](#)

[osteonecrosis of the jaw: A systematic review and Bayesian network meta-analysis - PubMed](#)

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## Diagnosis and Staging

Clinical diagnosis and radiography remain a solid diagnostic approach. Biomarkers for bone turnover and angiogenesis is still being explored

| Stage            | Description                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At risk category | Asymptomatic patients who have been treated with IV or oral antiresorptive therapy. No apparent necrotic bone.                                                                                                                                      |
| Stage 1          | Exposed and necrotic bone or fistula that probes to the bone in asymptomatic patients. There is no evidence of infection/inflammation.                                                                                                              |
| Stage 2          | Exposed and necrotic bone, or fistula that probes to the bone. Patients are symptomatic with clear signs of infection or inflammation.                                                                                                              |
| Stage 3          | Exposed and necrotic bone or fistulae that probes to the bone. Patients have pain, infection and one or more of the following: pathologic fracture, extra-oral fistula, osteolysis extending to the inferior border of the mandible or sinus floor. |

**Table 1 Staging and Treatment Strategies**

| <b>MRONJ† Staging</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Treatment Strategies‡</b>                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>At risk category</b> No apparent necrotic bone in patients who have been treated with either oral or IV bisphosphonates                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• No treatment indicated</li> <li>• Patient education</li> </ul>                                                                                                                                          |
| <b>Stage 0</b> No clinical evidence of necrotic bone, but non-specific clinical findings, radiographic changes and symptoms                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Systemic management, including the use of pain medication and antibiotics</li> </ul>                                                                                                                    |
| <b>Stage 1</b> Exposed and necrotic bone, or fistulae that probes to bone, in patients who are asymptomatic and have no evidence of infection                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>• Antibacterial mouth rinse</li> <li>• Clinical follow-up on a quarterly basis</li> <li>• Patient education and review of indications for continued bisphosphonate therapy</li> </ul>                     |
| <b>Stage 2</b> Exposed and necrotic bone, or fistulae that probes to bone, associated with infection as evidenced by pain and erythema in the region of the exposed bone with or without purulent drainage                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Symptomatic treatment with oral antibiotics</li> <li>• Oral antibacterial mouth rinse</li> <li>• Pain control</li> <li>• Debridement to relieve soft tissue irritation and infection control</li> </ul> |
| <b>Stage 3</b> Exposed and necrotic bone or a fistula that probes to bone in patients with pain, infection, and one or more of the following: exposed and necrotic bone extending beyond the region of alveolar bone,(i.e., inferior border and ramus in the mandible, maxillary sinus and zygoma in the maxilla) resulting in pathologic fracture, extra-oral fistula, oral antral/oral nasal communication, or osteolysis extending to the inferior border of the mandible of sinus floor | <ul style="list-style-type: none"> <li>• Antibacterial mouth rinse</li> <li>• Antibiotic therapy and pain control</li> <li>• Surgical debridement/resection for longer term palliation of infection and pain</li> </ul>                          |

† Exposed or probable bone in the maxillofacial region without resolution for greater than 8 weeks in patients treated with an antiresorptive and/or an antiangiogenic agent who have not received radiation therapy to the jaws.

‡ Regardless of the disease stage, mobile segments of bony sequestrum should be removed without exposing uninvolved bone. The extraction of symptomatic teeth within exposed, necrotic bone should be considered since it is unlikely that the extraction will exacerbate the established necrotic process.

## Resources

- [Medication-related osteonecrosis of the jaw: definition and best practice for prevention, diagnosis, and treatment - Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology \(oooojournal.net\).](#)
- [Preventing, identifying, and managing medication-related osteonecrosis of the jaw: a practical guide for nurses and other allied healthcare professionals | SpringerLink.](#)
- [Medication-related osteonecrosis of the jaw: Clinical and practical guidelines - PMC \(nih.gov\).](#)
- [American Association of Oral and Maxillofacial Surgeons' Position Paper on Medication-Related Osteonecrosis of the Jaws-2022 Update - PubMed \(nih.gov\).](#)
- [Medication-related osteonecrosis of the jaw in patients with cancer - UpToDate \(cnsu.edu\).](#)
- [Bisphosphonate Associated Osteonecrosis of the Jaw: An Update on Pathophysiology, Risk Factors, and Treatment \(hindawi.com\).](#)
- [Medication-Related Osteonecrosis of the Jaw: New Insights into Molecular Mechanisms and Cellular Therapeutic Approaches - PubMed \(nih.gov\).](#)
- [Medication-induced osteonecrosis of the jaw: a review of cases from the Food and Drug Administration Adverse Event Reporting System \(FAERS\) - PubMed \(nih.gov\)](#)
- [Incidence of medication-related osteonecrosis of the jaw and associated antiresorptive drugs in adult Finnish population - PubMed](#)